**U25AI055**

**Lab-2**

**Problem 1: Array Operations in C**

**Goal:**

**To understand the non-primitive data type Array and its operations. Write reusable functions for later use in the subject.**

**Tasks:**

**1. Array Creation: Create three signed integer arrays:**

**o SubArray1, SubArray2 of size 10.**

**o MainArray of size 20.**

**2. Operations: Implement a menu to perform the following operations on the arrays and**

**print the array elements after each operation:**

**o Insert elements at the end, beginning, and user-given position.**

**o Delete elements from the end, beginning, and user-given position.**

**o Reverse the array using:**

**▪ A temporary array to hold the result.**

**▪ In-place reversal (without a temporary array).**

**o Copy a sequence of elements from the subarray (SubArray1 or SubArray2) to**

**MainArray.**

**o Merge two sorted arrays in descending order.**

**o Display the sum of all array elements:**

**▪ Without recursion.**

**▪ Using recursion.**

**o Search the array for a user-given value.**

**o Separate the MainArray into sub-arrays for:**

**▪ Odd and even numbers.**

**▪ Positive and negative numbers.**

// 1: a: Array operations in C

```
#include <stdio.h>
#include <stdlib.h>
void readArray(int mainArray[], int size);
void printArray(int mainArray[], int size);
int insertElement(int mainArray[], int size, int element, int position);
int deleteElement(int mainArray[], int size, int position);
void reverseArray(int mainArray[], int size);
void reverseArrayInPlace(int mainArray[], int size);
int addElements(int mainArray[], int size);
int addElementsRecursively(int mainArray[], int size);
void searchElement(int mainArray[], int size, int element);
int main()
{
  int choice;
  int mainArray[20], subArray1[10], subArray2[10];
  int size;
  int element, position;
  printf("Enter number of elements: ");
  scanf("%d", &size);
  readArray(mainArray, size);
  printArray(mainArray, size);
  do
  {
    printf("\n1: Insert element at end\n");
    printf("2: Insert element at beginning\n");
    printf("3: Insert element at given position\n");
    printf("4: Delete last element\n");
    printf("5: Delete first element\n");
```

```
printf("6: Delete element at given position\n");
printf("7: Reverse array using temporary array\n");
printf("8: Reverse array without using temporary array\n");
printf("9: Display sum of all elements using recursion\n");
printf("10: Display sum of all elements without using recursion\n");
printf("11: Search element\n");
printf("Enter choice: ");
scanf("%d", &choice);
switch (choice)
{
  case 1:
    printf("Enter element: ");
    scanf("%d", &element);
    size = insertElement(mainArray, size, element, size);
    printArray(mainArray, size);
    break;
  case 2:
    printf("Enter element: ");
    scanf("%d", &element);
    size = insertElement(mainArray, size, element, 0);
    printArray(mainArray, size);
    break;
  case 3:
    printf("Enter element: ");
    scanf("%d", &element);
    printf("Enter position: ");
    scanf("%d", &position);
    size = insertElement(mainArray, size, element, position);
```

```
    printArray(mainArray, size);
    break;
  case 4:
    size = deleteElement(mainArray, size, size - 1);
    printArray(mainArray, size);
    break;
  case 5:
    size = deleteElement(mainArray, size, 0);
    printArray(mainArray, size);
    break;
  case 6:
    printf("Enter position: ");
    scanf("%d", &position);
    size = deleteElement(mainArray, size, position);
    printArray(mainArray, size);
    break;
  case 7:
    reverseArray(mainArray, size);
    printArray(mainArray, size);
    break;
  case 8:
    reverseArrayInPlace(mainArray, size);
    printArray(mainArray, size);
    break;
  case 9:
    printf("Sum of all elements is %d", addElements(mainArray, size));
    break;
  case 10:
```

```
            printf("Sum of all elements is %d", addElementsRecursively(mainArray, size));
            break;
        case 11:
            printf("Enter element to search: ");
            scanf("%d", &element);
            searchElement(mainArray, size, element);
            break;
        default:
            printf("Invalid choice");
    }
  } while (choice >= 1 && choice <= 11);
  return 0;
}
void readArray(int mainArray[], int size)
{
  printf("Enter elements of array: ");
  for (int i = 0; i < size; i++)
  {
    scanf("%d", &mainArray[i]);
  }
}
void printArray(int mainArray[], int size)
{
  printf("Elements of array are ");
  for (int i = 0; i < size; i++)
  {
    printf("%d\t", mainArray[i]);
  }
```

```
}
int insertElement(int mainArray[], int size, int element, int position)
{
   for (int i = size - 1; i >= position; i--)
   {
      mainArray[i + 1] = mainArray[i];
   }
   mainArray[position] = element;
   return size + 1;
}
int deleteElement(int mainArray[], int size, int position)
{
   for (int i = position; i < 20; i++)
   {
      mainArray[i] = mainArray[i + 1];
   }
   return size - 1;
}
void reverseArray(int mainArray[], int size)
{
   int temp[size];
   for (int i = 0; i < size; i++)
   {
      temp[size - i - 1] = mainArray[i];
   }
   for (int i = 0; i < size; i++)
   {
      mainArray[i] = temp [i];
```

```
    }
}
void reverseArrayInPlace(int mainArray[], int size)
{
    int left = 0, right = size - 1;
    while (left <= right)
    {
        int temp = mainArray[left];
        mainArray[left] = mainArray[right];
        mainArray[right] = temp;
        left++;
        right--;
    }
}
int addElements(int mainArray[], int size)
{
    int sum = 0;
    for (int i = 0; i < size; i++)
    {
        sum += mainArray[i];
    }
    return sum;
}
int addElementsRecursively(int mainArray[], int size)
{
    if (size == 0)
        return 0;
    return mainArray[size - 1] + addElementsRecursively(mainArray, size - 1);
```

```
}
void searchElement(int mainArray[], int size, int element)
{
   int position, flag = 0;
   for (int i = 0; i < size && flag == 0; i++)
   {
      if (mainArray[i] == element)
      {
         position = i;
         flag = 1;
      }
   }
   if (flag == 1)
   {
      printf("Element was found at %d\n", position);
   }
   else
   {
      printf("Element was not found\n");
   }
}
```

**Output:**

```
Enter number of elements: 6
Enter elements of array: 12
23
34
45
56
67
Elements of array are 12        23      34      45      56      67
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 1
Enter element: 78
Elements of array are 12        23      34      45      56      677
8
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 2
Enter element: 89
Elements of array are 89        12      23      34      45      566
7       78
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 3
Enter element: 90
Enter position: 2
Elements of array are 89        12      90      23      34      455
6       67      78
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 4
Elements of array are 89        12      90      23      34      455
6       67
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 5
Elements of array are 12        90      23      34      45      566
7
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 6
Enter position: 2
Elements of array are 12        90      34      45      56      67
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 7
Elements of array are 67        56      45      34      90      12
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 8
Elements of array are 12        90      34      45      56      67
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 9
Sum of all elements is 304
```

```
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 10
Sum of all elements is 304
1: Insert element at end
2: Insert element at beginning
3: Insert element at given position
4: Delete last element
5: Delete first element
6: Delete element at given position
7: Reverse array using temporary array
8: Reverse array without using temporary array
9: Display sum of all elements using recursion
10: Display sum of all elements without using recursion
11: Search element
Enter choice: 11
Enter element to search: 45
Element was found at 3
```

// 1: b: Array operations in C

```
#include <stdio.h>

#include <stdlib.h>

void readArray(int mainArray[], int size);
```

```
void printArray(int mainArray[], int size);
void copySubArray(int mainArray[], int subArray[]);
void mergeArrays(int mainArray[], int subArray1[], int subArray2[]);
void separateEvenAndOdd(int mainArray[], int size, int subArray1[], int subArray2[]);
void separatePositiveAndNegative(int mainArray[], int size, int subArray1[], int
subArray2[]);
int main()
{
  int choice;
  int mainArray[20], subArray1[10], subArray2[10];
  int size;
  do
  {
    printf("\n1: Copy sub array\n");
    printf("2: Merge two arrays\n");
    printf("3: Separate even and odd numbers\n");
    printf("4: Separate positive and negative numbers\n");
    printf("Enter choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
      case 1:
        printf("Enter elements of sub array: ");
        readArray(subArray1, 10);
        printf("Elements of sub array are ");
        printArray(subArray1, 10);
        copySubArray(mainArray, subArray1);
        printf("\nElements of main array are ");
```

```
    printArray(mainArray, 10);
    break;
  case 2:
    printf("Enter elements of sub array 1: ");
    readArray(subArray1, 10);
    printf("Elements of sub array 1 are ");
    printArray(subArray1, 10);
    printf("\nEnter elements of sub array 2: ");
    readArray(subArray2, 10);
    printf("Elements of sub array 2 are ");
    printArray(subArray2, 10);
    mergeArrays(mainArray, subArray1, subArray2);
    printf("\nElements of main array are ");
    printArray(mainArray, 20);
    break;
  case 3:
    printf("Enter number of elements: ");
    scanf("%d", &size);
    printf("Enter elements of main array: ");
    readArray(mainArray, size);
    printf("Elements of main array are ");
    printArray(mainArray, size);
    separateEvenAndOdd(mainArray, size, subArray1, subArray2);
    break;
  case 4:
    printf("Enter number of elements in main array: ");
    scanf("%d", &size);
    readArray(mainArray, size);
```

```
            printArray(mainArray, size);
            separatePositiveAndNegative(mainArray, size, subArray1, subArray2);
            break;
        default:
            printf("Invalid choice\n");
    }
  } while (choice >= 1 && choice <= 4);
  return 0;
}
void readArray(int mainArray[], int size)
{
  for (int i = 0; i < size; i++)
  {
    scanf("%d", &mainArray[i]);
  }
}
void printArray(int mainArray[], int size)
{
  for (int i = 0; i < size; i++)
  {
    printf("%d\t", mainArray[i]);
  }
}
void copySubArray(int mainArray[], int subArray[])
{
  for (int i = 0; i < 10; i++)
  {
    mainArray[i] = subArray[i];
```

```
    }
}
void mergeArrays(int mainArray[], int subArray1[], int subArray2[])
{
    int temp, j, i = 0;
    for (j = 0; j < 10; j++)
    {
        mainArray[i] = subArray1[j];
        i++;
    }
    for (j = 0; j < 10; j++)
    {
        mainArray[i] = subArray2[j];
        i++;
    }
    for (i = 0; i < 20; i++)
    {
        for (j = i + 1; j < 20; j++)
        {
            if (mainArray[i] < mainArray[j])
            {
                temp = mainArray[i];
                mainArray[i] = mainArray[j];
                mainArray[j] = temp;
            }
        }
    }
}
```

```
void separateEvenAndOdd(int mainArray[], int size, int subArray1[], int subArray2[])
{
   int j = 0, k = 0;
   for (int i = 0; i < size; i++)
   {
      if (mainArray[i] % 2 == 0)
      {
         subArray1[j] = mainArray[i];
         j++;
      }
      else
      {
         subArray2[k] = mainArray[i];
         k++;
      }
   }
   printf("\nEven numbers in sub array 1 are ");
   for (int i = 0; i < j; i++)
   {
      printf("%d\t", subArray1[i]);
   }
   printf("\nOdd numbers in sub array 2 are ");
   for (int i = 0; i < k; i++)
   {
      printf("%d\t", subArray2[i]);
   }
}
```

```
void separatePositiveAndNegative(int mainArray[], int size, int subArray1[], int
subArray2[])
{
   int j = 0, k = 0;
   for (int i = 0; i < size; i++)
   {
      if (mainArray[i] > 0)
      {
         subArray1[j] = mainArray[i];
         j++;
      }
      else if (mainArray[i] < 0)
      {
         subArray2[k] = mainArray[i];
         k++;
      }
   }
   printf("\nPositive numbers in sub array 1 are ");
   for (int i = 0; i < j; i++)
   {
      printf("%d\t", subArray1[i]);
   }
   printf("\nNegative numbers in sub array 2 are ");
   for (int i = 0; i < k; i++)
   {
      printf("%d\t", subArray2[i]);
   }
}
```

**Output:**

```
1: Copy sub array
2: Merge two arrays
3: Separate even and odd numbers
4: Separate positive and negative numbers
Enter choice: 1
Enter elements of sub array: 12
23
34
45
56
67
78
89
90
12
Elements of sub array are 12    23      34      45      56      677
8       89      90      12
Elements of main array are 12   23      34      45      56      677
8       89      90      12
```

```
Enter choice: 2
Enter elements of sub array 1: 12
23
34
45
56
67
78
89
90
12
Elements of sub array 1 are 12  23      34      45      56      677
8       89      90      12
Enter elements of sub array 2: 90
89
78
67
56
45
34
23
12
90
Elements of sub array 2 are 90  89      78      67      56      453
4       23      12      90
Elements of main array are 90   90      90      89      89      787
8       67      67      56      56      45      45      34      342
3       23      12      12      12
```

```
Enter choice: 3
Enter number of elements: 6
Enter elements of main array: 12
23
34
45
56
67
Elements of main array are 12   23      34      45      56      67
Even numbers in sub array 1 are 12      34      56
Odd numbers in sub array 2 are 23       45      67
```

```
Enter choice: 4
Enter number of elements in main array: 12
0
-23
34
45
56
67
-78
-89
-90
0
0
12
0       -23     34      45      56      67      -78     -89     -90
0       0       12
Positive numbers in sub array 1 are 34  45      56      67      12
Negative numbers in sub array 2 are -23 -78     -89     -90
```

**Problem 2: Design a Calendar**

**Goal:**

**Design a calendar for a given month and year using arrays to store:**

- **The names of the days of the week.**
- **The names of the months.**
- **The number of days in each month.**

**Task:**

- **Implement a C program to display the calendar for any given month and year.**

```
// 2: Design a calendar

#include <stdio.h>

int main()

{

        int year, month;

  char daysOfWeek[7][10] = {"Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"};

  char monthsInYear[12][10] = {"January", "February", "March", "April", "May", "June",
"July", "August", "September", "October", "November", "December"};

  printf("Enter year: ");
```

```
        scanf("%d", &year);
printf("Enter month: ");
        scanf("%d", &month);
        int reference = 1990, totalDays = 0;
        for (int i = reference; i < year; i++)
        {
                if (i % 4 == 0)
  {
    totalDays += 366;
  }
                else
  {
    totalDays += 365;
  }
        }
        for (int i = 1; i < month; i++)
        {
                switch (i)
                {
                        case 2:
                                if (year % 4 == 0)
        {
          totalDays += 29;
        }
                                else
        {
          totalDays += 28;
        }
```

```
                        break;
                    case 1: case 3: case 5: case 7: case 8: case 10: case 12:
                        totalDays += 31;
                        break;
                    default:
                        totalDays += 30;
                }
        }
        int start = totalDays % 7, row, column, date = 1, days;
        switch (month)
        {
                case 2:
                    if (year % 4 == 0)
      {
         days = 29;
      }
                    else
      {
         days = 28;
      }
                    break;
                case 1: case 3: case 5: case 7: case 8: case 10: case 12:
                    days = 31;
                    break;
                default:
                    days = 30;
        }
printf("%s, %d\n", monthsInYear[month - 1], year);
```

```
    for (int i = 0; i < 7; i++)
    {
        printf("%s\t", daysOfWeek[i]);
    }
    printf("\n");
        for (column = 0; column < 7 && date <= days; column++)
        {
            if (column < start)
                printf("   \t");
            else
            {
                printf("%3d\t", date);
                date++;
            }
        }
        printf("\n");
        for (row = 1; date <= days; row++)
        {
            for (column = 0; column < 7 && date <= days; column++)
            {
                printf("%3d\t", date);
                date++;
            }
            printf("\n");
        }
        return 0;
}
```

**Output:**

```
Enter year: 2026
Enter month: 1
January, 2026
Mon     Tue     Wed     Thu     Fri     Sat     Sun
                         1       2       3       4
  5       6       7       8       9      10      11
 12      13      14      15      16      17      18
 19      20      21      22      23      24      25
 26      27      28      29      30      31
```